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compact1, compact2, compact3

java.util.concurrent.atomic

Class LongAdder

java.lang.Object
    java.lang.Number
        java.util.concurrent.atomic.LongAdder

All Implemented Interfaces:

Serializable

```
public class LongAdder
extends Number
implements Serializable
```

One or more variables that together maintain an initially zero `long` sum. When updates (method `add(long)`) are contended across threads, the set of variables may grow dynamically to reduce contention. Method `sum()` (or, equivalently, `longValue()`) returns the current total combined across the variables maintaining the sum.

This class is usually preferable to `AtomicLong` when multiple threads update a common sum that is used for purposes such as collecting statistics, not for fine-grained synchronization control. Under low update contention, the two classes have similar characteristics. But under high contention, expected throughput of this class is significantly higher, at the expense of higher space consumption.

LongAdders can be used with a `ConcurrentHashMap` to maintain a scalable frequency map (a form of histogram or multiset). For example, to add a count to a `ConcurrentHashMap<String,LongAdder> freqs`, initializing if not already present, you can use `freqs.computeIfAbsent(k -> new LongAdder()).increment();`

This class extends `Number`, but does *not* define methods such as `equals`, `hashCode` and `compareTo` because instances are expected to be mutated, and so are not useful as collection keys.

Since:

1.8

See Also:

Serialized Form

Constructor Summary

Constructors

Constructor	Description
`LongAdder()`	Creates a new adder with initial sum of zero.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
`void`	`add(long x)`	Adds the given value.
`void`	`decrement()`	Equivalent to `add(-1)`.
`double`	`doubleValue()`	Returns the `sum()` as a `double` after a widening primitive conversion.
`float`	`floatValue()`	Returns the `sum()` as a `float` after a widening primitive conversion.
`void`	`increment()`	Equivalent to `add(1)`.
`int`	`intValue()`	Returns the `sum()` as an `int` after a narrowing primitive conversion.
`long`	`longValue()`	Equivalent to `sum()`.
`void`	`reset()`	Resets variables maintaining the sum to zero.
`long`	`sum()`	Returns the current sum.
`long`	`sumThenReset()`	Equivalent in effect to `sum()` followed by `reset()`.
`String`	`toString()`	Returns the String representation of the `sum()`.

Methods inherited from class java.lang.Number

byteValue, shortValue

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, wait, wait, wait

Constructor Detail

LongAdder

```
public LongAdder()
```

Creates a new adder with initial sum of zero.

Method Detail

add

```
public void add(long x)
```

Adds the given value.

Parameters:

`x - the value to add`

increment

```
public void increment()
```

Equivalent to `add(1)`.

decrement

```
public void decrement()
```

Equivalent to `add(-1)`.

sum

```
public long sum()
```

Returns the current sum. The returned value is *NOT* an atomic snapshot; invocation in the absence of concurrent updates returns an accurate result, but concurrent updates that occur while the sum is being calculated might not be incorporated.

Returns:

the sum

reset

```
public void reset()
```

Resets variables maintaining the sum to zero. This method may be a useful alternative to creating a new adder, but is only effective if there are no concurrent updates. Because this method is intrinsically racy, it should only be used when it is known that no threads are concurrently updating.

sumThenReset

```
public long sumThenReset()
```

Equivalent in effect to `sum()` followed by `reset()`. This method may apply for example during quiescent points between multithreaded computations. If there are updates concurrent with this method, the returned value is *not* guaranteed to be the final value occurring before the reset.

Returns:

the sum

toString

```
public String toString()
```

Returns the String representation of the `sum()`.

Overrides:

`toString` in class `Object`

Returns:

the String representation of the `sum()`

longValue

```
public long longValue()
```

Equivalent to `sum()`.

Specified by:

`longValue` in class `Number`

Returns:

the sum

intValue

```
public int intValue()
```

Returns the `sum()` as an `int` after a narrowing primitive conversion.

Specified by:

`intValue` in class `Number`

Returns:

the numeric value represented by this object after conversion to type int.

floatValue

```
public float floatValue()
```

Returns the `sum()` as a `float` after a widening primitive conversion.

Specified by:

`floatValue` in class `Number`

Returns:

the numeric value represented by this object after conversion to type float.

doubleValue

```
public double doubleValue()
```

Returns the `sum()` as a `double` after a widening primitive conversion.

Specified by:

`doubleValue` in class `Number`

Returns:

```
the numeric value represented by this object after conversion to
type double.
```


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